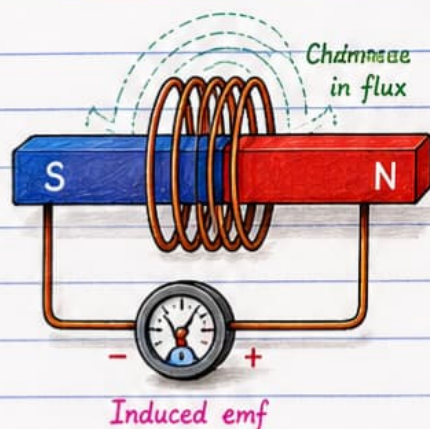


ELECTROMAGNETIC INDUCTION



"Whenever there is a change in magnetic flux linked with a conductor, an emf is induced in it."



1. Electromagnetic Induction :

- The phenomenon of inducing an emf in a conductor due to change in magnetic flux linked with it is called **electromagnetic induction**.
- It was discovered by Michael Faraday in 1831.
- This is the basis of generators, transformers and many electrical devices.

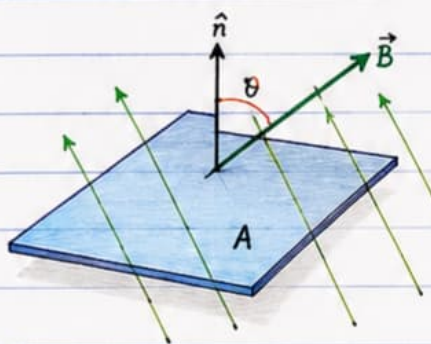
Key Idea

No change in flux $\Rightarrow$ No emf induced
Change in flux $\Rightarrow$ emf induced

2. Magnetic Flux (Φ):

- Magnetic flux through a surface is the total number of magnetic field lines passing through it.
 - Mathematically, $\Phi = B \cdot A \cos \theta$
- where,
 B = Magnetic field (tesla)
 A = Area of the surface (m^2)
 θ = Angle between B and normal to the surface

Magnetic Flux Through a Surface



- $\theta = 0^\circ$ $\Phi = BA$ (Maximum)
- $\theta = 90^\circ$ $\Phi = 0$ (Minimum)
- General case
 $\Phi = BA \cos \theta$

3. Faraday's Law of Electromagnetic Induction :

The magnitude of induced emf in a circuit is equal to the rate of change of magnetic flux linked with it.

$$\mathcal{E} = -N \frac{d\Phi}{dt}$$

where,
 $\mathcal{E}$ = Induced emf (volt)
 N = Number of turns in the coil
 $\frac{d\Phi}{dt}$ = Rate of change of magnetic flux (weber/second)
 The negative sign shows the direction of emf opposes the change in flux (Lenz's Law).

Meaning of Negative Sign (Lenz's Law)

The induced emf always acts in such a direction that it opposes the cause which produces it.

Example : If flux increases, induced emf produces current that opposes the increase.

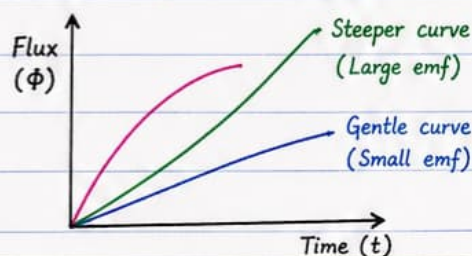
Unit Check

1 weber (Wb) = 1 tesla $\times$ 1 m^2
 1 volt = 1 weber/second

4. Factors Affecting Induced emf :

- Rate of change of flux $\left(\frac{d\Phi}{dt}\right) \rightarrow$ Faster change, larger emf.
- Number of turns (N) $\rightarrow$ More turns, larger emf.
- Strength of magnetic field (B) $\rightarrow$ Stronger field, larger emf.
- Area of the coil (A) $\rightarrow$ Larger area, larger flux change.
- Orientation (θ) $\rightarrow$ Maximum when $\theta = 0^\circ$.

More Change $\Rightarrow$ More emf



ELECTROMAGNETIC INDUCTION

5. Motional emf :

- When a conductor moves in a magnetic field such that it cuts magnetic field lines, an emf is induced in it.
- The direction of induced emf is given by Fleming's Right Hand Rule.

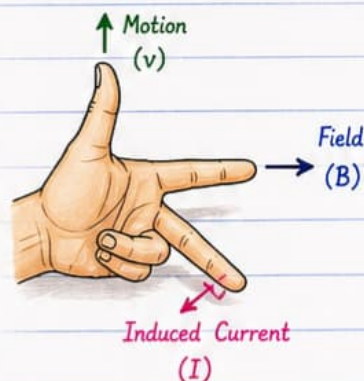
Fleming's Right Hand Rule

Stretch the thumb, forefinger and middle finger of your right hand mutually perpendicular to each other.

Forefinger → Magnetic Field (B)

Thumb → Motion or Velocity (v)

Middle finger → Induced Current (I)



6. Induced emf in a Straight Conductor Moving in a Uniform Magnetic Field :

- Consider a conductor of length l moving with velocity v in a uniform magnetic field B perpendicular to both v and B .
- Charges inside the conductor experience magnetic force and get separated.
- An emf is induced between the ends.

Magnitude of induced emf :

$$\mathcal{E} = B l v$$

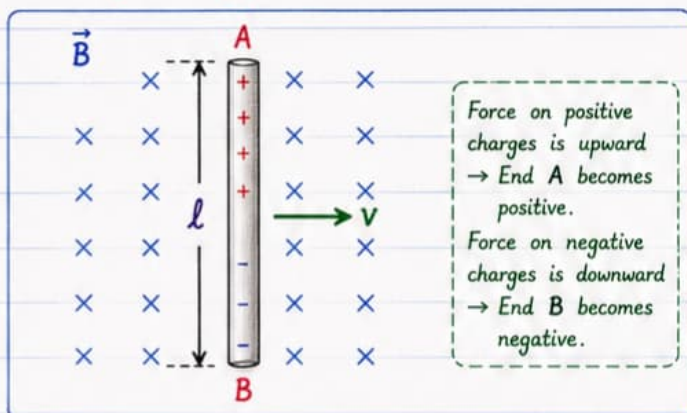
where,

B = Magnetic field (tesla)

l = Length of the conductor (m)

v = Velocity of the conductor (m/s)

$\mathcal{E}$ = Induced emf (volt)



If the conductor is moved in the opposite direction, the polarity of ends will reverse.

7. Induced emf in a Rotating Coil :

- A rectangular coil of N turns, area A , rotates with angular speed ω in a uniform magnetic field B .
- The magnetic flux linked with the coil at any instant is,

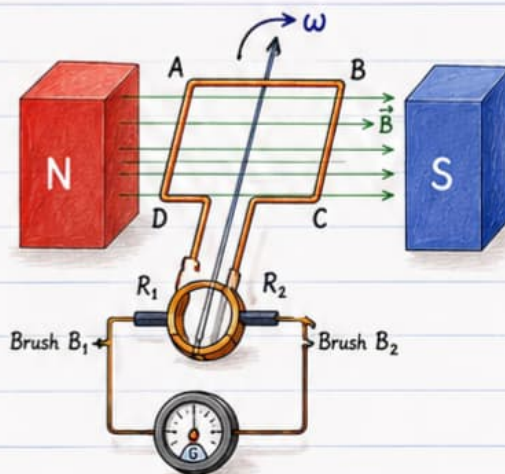
$$\Phi = NBA \cos \theta$$

- Using Faraday's Law,

$$\mathcal{E} = -N \frac{d\Phi}{dt} = NBA \omega \sin \theta$$

- The induced emf varies sinusoidally with time.

$$\mathcal{E} = \mathcal{E}_0 \sin \omega t, \text{ where } \mathcal{E}_0 = NBA \omega$$



$\mathcal{E}$ is maximum when $\theta = 90^\circ$

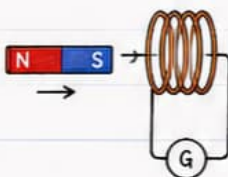
$\mathcal{E}$ is zero when $\theta = 0^\circ$ or 180°

8. Lenz's Law (Again) :

The direction of induced emf (and current) is such that it opposes the change in magnetic flux that produces it.

Example :

If a magnet is pushed towards a coil, the coil produces an induced current which creates a magnetic field to repel the approaching magnet.



QUICK SUMMARY

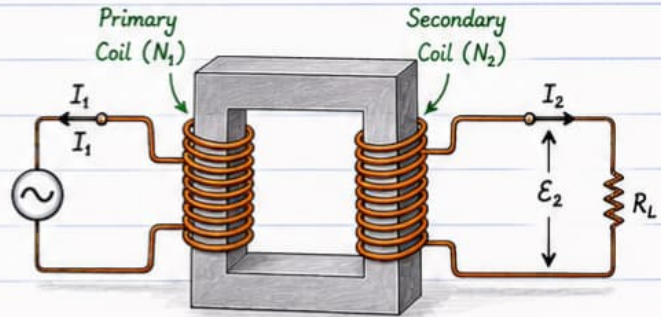
Induced emf	Produced due to change in magnetic flux.
Faraday's Law	$\mathcal{E} = -N \frac{d\Phi}{dt}$
Lenz's Law	Induced emf opposes the change in flux.
Motional emf	$\mathcal{E} = B l v$ (when conductor cuts field lines)
Rotating coil emf	$\mathcal{E} = NBA \omega \sin \omega t$
Direction	Fleming's Right Hand Rule

★ Remember : No change in flux → No induced emf ★

ELECTROMAGNETIC INDUCTION

9. Mutual Induction :

- When change of current in a coil produces emf in a nearby coil, the phenomenon is called **mutual induction**.
- It is the basis of transformer.



Key Point

- Change in current in primary coil (I_1)
 $\Rightarrow$ Change in magnetic flux in core
 $\Rightarrow$ Induced emf in secondary coil (E_2)

Formula

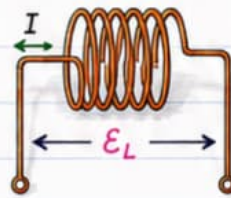
$$E_2 = -M \frac{dI_1}{dt}$$

where,

- M = Mutual inductance (henry)
- I_1 = Current in primary coil (ampere)
- E_2 = Induced emf in secondary (volt)

10. Self Induction :

- When change of current in a coil induces an emf in the same coil, the phenomenon is called **self induction**.
- The induced emf in this case opposes the change in current.



Self Induced emf

$$E_L = -L \frac{dI}{dt}$$

where,

- L = Self inductance of the coil (H)
- I = Current in the coil (A)
- E_L = Self induced emf (V)

Self Inductance (L) depends on :

- Number of turns (N) $\rightarrow$ More turns, larger L
- Area of cross-section (A) $\rightarrow$ Larger area, larger L
- Length of the coil (ℓ) $\rightarrow$ Longer coil, smaller L
- Nature of core $\rightarrow$ Iron core $\Rightarrow$ Larger L

Unit Check

$$\begin{aligned} 1 \text{ henry (H)} &= 1 \frac{\text{Wb}}{\text{A}} \\ 1 \text{ mH} &= 10^{-3} \text{ H} \\ 1 \mu\text{H} &= 10^{-6} \text{ H} \end{aligned}$$

Lenz's Law in Self Induction

The self-induced emf always opposes the change in current producing it. Hence, it resists the change in current.

11. Energy Consideration in Induction :

- When an emf is induced, current flows in the circuit.
- Work is done against this induced emf.
- This work appears as electrical energy in the circuit.
- Hence, energy is conserved.

Energy stored in an inductor (or a coil)

$$U = \frac{1}{2} L I^2$$

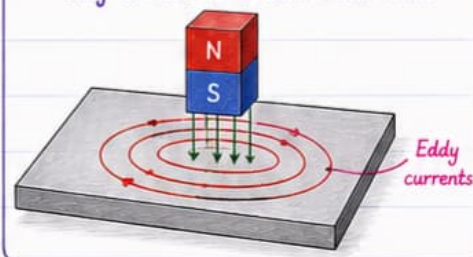
where,

- U = Energy stored (joule)
- L = Inductance (henry)
- I = Current (ampere)

12. Eddy Currents :

- When a conductor moves in a magnetic field or is placed in a changing magnetic field, **circulating currents** are set up inside the bulk of conductor.
- These are called **eddy currents**.
- Eddy currents cause loss of energy by heating.
- To reduce eddy currents, cores are made up of thin laminated sheets.

Eddy Currents in a Solid Metal Plate



Applications

- ★ Electromagnetic brakes
- ★ Damping in moving coil instruments
- ★ Induction furnaces

IMPORTANT POINTS (Quick Revision)

- ★ Induced emf exists only when there is change in magnetic flux.
- ★ Direction of induced emf is given by Lenz's Law / Fleming's Right Hand Rule.

- ★ Induced emf in straight conductor $E = B \ell v$ (when $v \perp B$)
- ★ Induced emf in rotating coil $E = N B A \omega \sin \theta$

- ★ Self induced emf $E_L = -L \frac{dI}{dt}$
- ★ Energy stored in inductor $U = \frac{1}{2} L I^2$



Change in Magnetic Flux
 $\Rightarrow$ Induced EMF

★ Remember : No change in flux $\Rightarrow$ No induced emf ★

ELECTROMAGNETIC INDUCTION

13. Direction of Induced Current (Summary):

Lenz's Law

Nature never allows the cause which produces it.

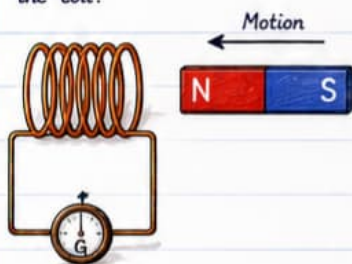
So, induced current always flows in such a direction that it opposes the change in magnetic flux that produces it.

Change in Flux (Φ)	Induced Current	Reason
Φ Increasing	Opposes the increase (create opposite field)	To reduce the increase in flux
Φ Decreasing	Supports the decrease (create same field)	To oppose the decrease in flux
Φ Constant	No induced current	No change in flux

14. Important Examples :

(i) Moving Magnet and Coil

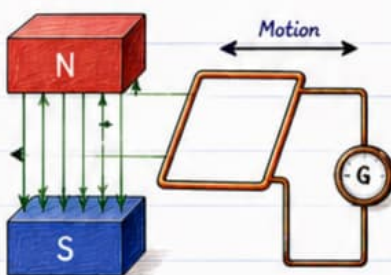
When a magnet is moved towards a coil, induced current flows in the coil.



If N-pole moves towards the coil, the face of the coil towards the magnet becomes **N-pole**.
(Opposes the approach)

(ii) Moving Coil in Magnetic Field

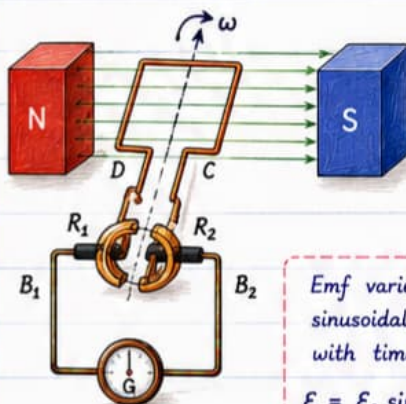
When a coil is moved in a magnetic field, induced current flows.



Direction is given by
Fleming's Right Hand Rule
or **Lenz's Law**.

(iii) Rotating Coil in Magnetic Field

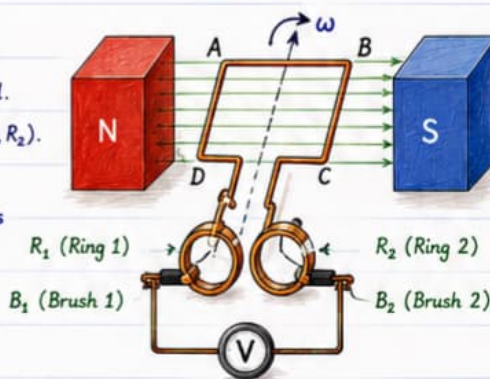
A coil rotating in uniform magnetic field produces alternating emf and current.



Emf varies sinusoidally with time.
 $\mathcal{E} = \mathcal{E}_0 \sin \omega t$

15. AC Generator (Basic Principle):

- A rectangular coil rotates in a uniform magnetic field.
- The ends of the coil are connected to two slip rings (R_1, R_2).
- Two carbon brushes (B_1, B_2) press on the rings.
- As the coil rotates, emf is induced and appears across the brushes.
- The output is alternating current (AC).
- Frequency, $f = \frac{N}{60}$ (N = speed in rpm)



Peak (Maximum) Emf
 $\mathcal{E}_0 = NBA\omega$

Instantaneous Emf
 $\mathcal{E} = \mathcal{E}_0 \sin \omega t$

where,

N = Number of turns

B = Magnetic field

A = Area of coil

ω = Angular speed

Applications of Electromagnetic Induction

- ★ Electric generators
- ★ Transformers
- ★ Induction motors
- ★ Induction cooktops
- ★ Metal detectors
- ★ Wireless charging
- ★ Mic (Dynamic) & Guitar pickups
- ★ Eddy current brakes



Compare : Self Induction vs Mutual Induction

Point	Self Induction	Mutual Induction
Definition	Induced emf in the same coil	Induced emf in another coil
Cause	Change in current in same coil	Change in current in nearby coil
Symbol	L (Self inductance)	M (Mutual inductance)
Emf	$\mathcal{E}_L = -L \frac{dI}{dt}$	$\mathcal{E} = -M \frac{dI}{dt}$
Example	Self induction in inductor	Transformer

Common Mistakes

- X Ignoring the sign in Faraday's Law.
- X Confusing cause and effect in Lenz's Law.
- X Forgetting that induced emf exists only when there is change in flux.
- X Using wrong hand rule.

Key Takeaway



Change in magnetic flux $\rightarrow$ Induced EMF
Direction $\rightarrow$ Opposes the change (Lenz's Law)

$\therefore$ Concept Mastered ☒